

What Is a Mechanism?

An Atlas of Mechanistic Views for Interpretability Research

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Abstract

Mechanistic interpretability has produced circuits, features, causal subspaces, and activation-steered algorithms. Each result rests on an implicit answer to a prior question: what kind of thing is a mechanism? A head-level component, a functional role, a causal subspace, a gauge-invariant structure, and a training-time trajectory carry different identity criteria, require different evidence, and license different inferences. This is not a terminological inconvenience—it means that many apparent disagreements in the literature are ontological disagreements being argued at the wrong level.

This paper introduces *mechanistic views*: explicit background accounts of what mechanisms are, how they are identified, when two descriptions refer to the same mechanism, and what evidence can support claims about them. A view is a tuple $(O, \sim, E, F, T)$ where O is a set of mechanism candidates, $\sim$ is an equivalence relation on O specifying identity, E is an evidential standard, F is the appropriate formalism, and T is the explanatory target. We state coherence conditions on the tuple and prove that incoherent views produce systematically unsatisfiable evidential demands.

We organize the space of possible views along five axes and present an atlas of eight views currently operative in the interpretability literature: object, role, subspace, structural, process, stratified, perspectival, and instrumental. For each view we give the full axis assignments, characteristic evidence types, failure modes, and the experiments that would discriminate it from neighboring views. We then develop three worked examples at depth—IOI, induction heads, and SAE features—showing what each view claims, what it licenses, and what evidence remains open. Finally, we identify five recurring decision points in empirical work and explain what each view implies for them.

The framework is not a proposal for a single correct ontology. It is a diagnostic tool: making the view explicit lets researchers state claims at the level their evidence supports, locate disagreements at their source, and design experiments that discriminate views rather than confirming them.

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*Working draft, June 2026. Companion to Tower (2026a) *Mechanistic Validity* and Tower (2026b) *The Geometry of Mechanisms*.

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1 Introduction

Mechanistic interpretability aims to explain what trained neural networks are computing. The field has made this aim concrete: transformer circuits provide a component-level vocabulary for attention heads and MLPs [Elhage et al., 2021]; induction heads account for a mechanism involved in in-context learning [Olsson et al., 2022]; the indirect object identification (IOI) circuit names the attention heads involved in a specific syntactic task [Wang et al., 2022]; sparse autoencoders recover interpretable directions from activation spaces [Bricken et al., 2023, Templeton et al., 2024]; causal abstraction methods test whether high-level variables are realized in distributed representations [Geiger et al., 2021, 2024]; and weight-space analyses recover circuit structure without forward passes [Tower, 2026c].

Yet the word *mechanism* does not mean the same thing in all of these cases. In one paper a mechanism is a set of attention heads; in another it is a functional role such as “name mover”; in another it is a causal subspace recovered by distributed alignment search; in another it is a training-time transition. These are not merely terminological differences. They carry different answers to four basic questions:

- (1) **Existence:** what objects count as mechanisms?
- (2) **Identity:** when do two descriptions refer to the same mechanism?
- (3) **Evidence:** what measurements can warrant a claim about a mechanism?
- (4) **Explanation:** what has been achieved when a mechanism is found?

Most published work answers these questions implicitly. The answers are recoverable from the methods used and the conclusions drawn, but they are not stated—and because they are not stated, disagreements that are fundamentally ontological are argued as disagreements about experimental design.

Three motivating cases. Underdetermination. Activation patching establishes that a head is causally relevant to a behavior. Does this establish a mechanism? Under the object view, perhaps yes. Under the role view, only if the functional role is independently specified and tested. Under the subspace view, not unless the relevant variable is localized to a specific causal subspace. Under the structural view, not unless the result is invariant under the relevant parameter symmetries. A single activation-patching result means different things under different views, and the disagreement cannot be resolved by further patching experiments alone—because all such experiments are equally consistent with multiple views.

Cross-model comparison. The claim that GPT-2 Small and Pythia-160M implement the same induction head mechanism requires a notion of mechanism identity that is cross-model. Component indices are trivially not cross-model. Functional roles may be, if independently specified. Gauge-invariant subspace structure may be, if a canonical transport map is defined. Training-time trajectories may be, if the dynamical basins correspond. Which notion is operative determines whether “the same mechanism” is meaningful across architectures—and therefore whether cross-model generalization claims are well-formed.

Safety-relevant inference. Safety work reasons from “circuit X is present” to “the model has goal Y ” or “the model would behave Z under distribution shift.” These inferences require different things depending on what a mechanism is. Under an object view, circuit presence is a

fact about specific components in a specific model on a specific distribution. Under a structural view, a circuit present as a gauge-invariant structure is robust to reparameterization. Under the instrumental view, a circuit label is warranted by behavioral predictions. The inference from circuit presence to safety-relevant conclusion requires specifying which of these is operative.

Contributions. This paper makes the following contributions:

1. A formal definition of a *mechanistic view* as a five-component structure $(O, \sim, E, F, T)$ with typed components and coherence conditions stated as propositions (§2).
2. An atlas of eight views currently operative in the interpretability literature, with full axis assignments, evidence standards, discriminating experiments, and failure modes (§3).
3. Three worked examples—IOI, induction heads, SAE features—showing what each view claims, what it licenses, and what evidence is needed to advance from one view to another (§4).
4. An analysis of five recurring decision points in empirical work and what each view implies for them (§5).
5. Positioning of Mechanistic Views relative to the companion frameworks of Mechanistic Validity [Tower, 2026a] and the Geometry of Mechanisms [Tower, 2026b], with explicit identification of what each framework presupposes and what it provides (§6).

What this paper is not. It is not a proposal to replace the eight views with a single preferred ontology. Each view is appropriate for some questions and inappropriate for others. The goal is to make view choice explicit so that it can be justified or revised, not to eliminate plurality. It is also not a paper about methodology per se: methods (patching, DAS, SAEs, weight-space analysis) are discussed only in their relation to views, not evaluated on their own technical merits.

Relationship to companion papers. MechVal [Tower, 2026a] provides criteria for evaluating whether a mechanistic claim is warranted by evidence. Those criteria presuppose that the claim type is fixed. The Geometry of Mechanisms [Tower, 2026b] provides the formal mathematical objects—Grassmannian structural causal models, parallel transport, holonomy, sheaf cohomology—that the subspace and structural views implicitly require. That formalism presupposes that the subspace view is appropriate. The present paper asks what makes those presuppositions explicit and justifiable.

2 Mechanistic Views

2.1 Formal Definition

A mechanistic view is a background commitment that a researcher makes, explicitly or implicitly, when they ask what a mechanism is. We give a formal definition, then unpack each component.

Definition 2.1 (Mechanistic view). *A mechanistic view σ is a tuple $(O, \sim, E, F, T)$ where:*

- (i) O is a set, the **ontology**, whose elements are the entities that count as mechanisms under σ .
- (ii) $\sim \subseteq O \times O$ is an equivalence relation on O , the **identity criterion**: $x \sim y$ means descriptions x and y refer to the same mechanism under σ .
- (iii) E is a set of **evidential standards**: pairs (m, C) where m is a measurement type and $C \subseteq 2^O / \sim$ is the set of claims about equivalence classes of mechanisms that m can in principle support.
- (iv) F is a **formalism**: a mathematical or representational language in which elements of O can be expressed and $\sim$ can be computed.
- (v) T is a **target**: a specification of the phenomena the view is designed to explain.

Several features of Definition 2.1 deserve comment.

The identity criterion $\sim$ is required to be an equivalence relation (reflexive, symmetric, transitive). This is not automatic: some informal notions of mechanism identity in the literature are not transitive. For example, saying mechanisms A and B are “related” by shared components, and B and C are “related” by shared function, does not imply A and C are the same mechanism. Making $\sim$ an equivalence relation forces precision.

The evidential standard E is explicitly typed: each measurement type m is associated with the claims C it can support. This rules out the common practice of using a measurement to support a claim that is not in the range of that measurement. Activation patching establishes causal relevance of a component; it is not in the range of subspace localization claims unless supplemented.

Definition 2.2 (Coherence). *A view $\sigma = (O, \sim, E, F, T)$ is coherent if:*

- (i) **Identity is grounded**: $\sim$ is definable using only the objects in O and the language of F . (A view whose identity criterion involves objects not in its own ontology is incoherent.)
- (ii) **Evidence tracks identity**: for every measurement type $m \in E$, the claims $C(m)$ are expressible in terms of $\sim$ -classes, not finer-grained distinctions. (Evidence that distinguishes objects within a $\sim$ -class is picking up on a different ontology than the one declared.)
- (iii) **Formalism is expressive**: every element of O has a representation in F , and $\sim$ is computable in F .

Proposition 2.1 (Incoherent views produce contradictory demands). *Let $\sigma = (O, \sim, E, F, T)$ be a view that violates coherence condition (ii): there exists a measurement type $m \in E$ and objects $x, y \in O$ with $x \sim y$ but $m(x) \neq m(y)$. Then there exist mechanistic claims that are simultaneously required by $\sim$ to be true (since x and y are the same mechanism) and required by E to be false (since m distinguishes them). No experiment can satisfy both demands.*

Proof. Let ϕ be the claim “ x and y are the same mechanism.” Since $x \sim y$, ϕ is true under σ . The measurement m produces $m(x) \neq m(y)$. If m is an evidence type for mechanism identity (i.e., different m -values imply different mechanisms), then $m(x) \neq m(y)$ implies ϕ is false. The view asserts ϕ and $\neg\phi$ simultaneously. $\square$ $\square$

Proposition 2.1 is useful diagnostically. A common pattern in interpretability is to declare that mechanisms are gauge-invariant structures (structural view ontology) but then to use activation-patching results as evidence of mechanism presence without checking gauge invariance. If the patching result changes under a reparameterization that preserves computation, it is distinguishing objects that the structural view says are identical.

2.2 The Five Axes

We organize the components of Definition 2.1 into five named axes. The axes are not independent: as coherence requires, choices on one axis constrain the others.

Ontology (O). What kind of entity counts as a mechanism? Options include: concrete model components (heads, neurons, MLP sublayers, SAE dictionary elements), functional roles (descriptions of what a component does independently of which component does it), causal subspaces (linear or nonlinear submanifolds of the residual stream that mediate a causal variable), gauge-invariant structures (equivalence classes of weight configurations under symmetries that preserve the computation), temporally extended processes (formation pathways or execution trajectories), points in a stratified mechanism space, measurement perspectives, and predictive models.

Identity ($\sim$). When do two mechanism descriptions denote the same mechanism? Options include: component overlap ($x \sim y$ iff they name the same set of heads or neurons), role equivalence ($x \sim y$ iff they specify the same functional input–output mapping), subspace proximity ($x \sim y$ iff the principal angles between the causal subspaces are all below a threshold θ , i.e. geodesic distance on $\text{Gr}(k, d)$ is small), gauge-orbit membership ($x \sim y$ iff there exists a computation-preserving symmetry g with $g \cdot x = y$), and dynamical-basin membership ($x \sim y$ iff the formation trajectories converge to the same attractor).

Evidence (E). What measurements can support what claims? Key distinctions: activation patching establishes causal relevance of components under a fixed distribution; DAS/IIA with linear alignment constraints measures causal subspace identity; weight composition scores measure structural information flow independent of any input distribution; checkpoint analysis establishes formation dynamics; multi-method robustness tests perspectival convergence.

Formalism (F). What representational language is appropriate? Directed graphs (circuits); functional decompositions (roles); Grassmannian geometry $\text{Gr}(k, d)$ (subspaces); fiber bundles and gauge groups (structural view); dynamical systems and phase portraits (process view); Whitney stratifications (stratified view); measurement algebras (perspectival view).

Target (T). What is the mechanism supposed to explain? A specific behavioral output on a specific distribution; a functional class across models; a representational variable across inputs; a computation class across parameterizations; a formation event; a resolution-relative phenomenon; or a behavioral prediction.

3 An Atlas of Views

We identify eight views currently operative in the literature. Table 1 gives the full axis assignments; Figure 1 shows how the five axes organize the eight views. The views are not mutually exclusive: a single paper may use several, and convergence across views constitutes stronger evidence than any single view. The point is to distinguish them so that each view’s commitments and limitations can be assessed independently.

View	Ontology (O)	Identity ($\sim$)	Evidence (E)	Formalism (F)	Target (T)
Object	Concrete part: head, neuron, feature	Component overlap	Ablation, activation patching	Directed graph	Specific behavior
Role	Functional role in computation	Role equivalence (same I/O mapping)	Role-specific causal tests, cross-model	Functional decomposition	Functional class
Subspace	Causal subspace of residual stream	Same projector; $d_{Gr} < \theta$	DAS/IIA (linear), subspace stability	$Gr(k, d)$	Representational variable
Structural	Gauge-invariant relational structure	Gauge-orbit membership	Holonomy, comp. scores, reparameterization	Fiber bundle quotient	Computation class
Process	Formation trajectory or execution regime	Same trajectory type or dynamical basin	Training checkpoints, formation knockouts	Dynamical system	Mechanism origin
Stratified	Point in stratified mechanism space	Same stratum + local equivalence	Dimensionality tests, localization diagnostics	Whitney stratification	Resolution-relative phenomenon
Perspectival	Projection of mechanism onto context	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative observation
Instrumental	Useful predictive or interventional model	Predictive equivalence	Forecast accuracy, intervention utility	Model theory	Behavioral prediction

Table 1: The eight mechanistic views with full axis assignments.

3.1 Object View

The object view treats mechanisms as concrete model components: specific attention heads, neurons, MLP sublayers, or sparse autoencoder dictionary elements. It is the implicit view in circuit diagrams. The IOI circuit as reported by Wang et al. [2022]—26 attention heads organized into functional groups—is an object-level description: the claim is about those specific computational units.

Identity. Two circuit descriptions refer to the same mechanism iff they name the same components. This is precise within a model but trivially fails cross-model: a head in GPT-2

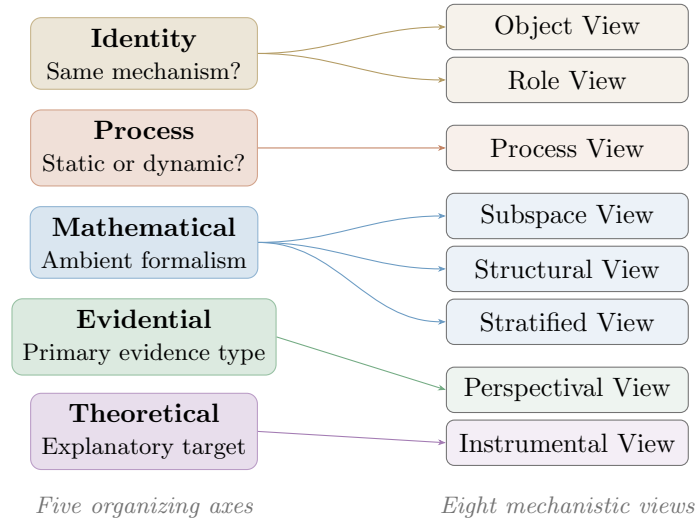


Figure 1: The five axes of variation (left) and the eight mechanistic views they organize (right). Each axis addresses one component of the formal definition (Definition 2.1). The three mathematical-family views—subspace, structural, and stratified—share a common ambient formalism but differ in what they treat as the identity criterion and the relevant quotient.

Small is not an object in Pythia-160M.

Evidence. Ablation and activation patching are the natural evidence types. Removing a component’s activation and observing behavioral degradation establishes causal necessity; replacing it with a counterfactual activation establishes causal sufficiency relative to that counterfactual. The method is strongest when the component set is stable across prompt distributions and the ablation effect is large and specific.

Failure modes. *Role inflation:* a component is identified by patching, then labeled with a role (“name mover,” “induction head”) that implies richer mechanistic content than the object-level evidence supports. The component identity is established; the role is a hypothesis. *Distribution sensitivity:* the circuit identified by patching on one distribution may differ substantially on another, meaning the claimed object-level mechanism is distribution-relative.

Discriminating experiment. To test whether the object view is appropriate: ablate the identified component set and test whether behavioral degradation is (a) specific to the behavior the mechanism is supposed to implement and (b) stable across prompt distributions. If (b) fails, the object-level description is distribution-relative and a role or subspace description may be more appropriate.

3.2 Role View

The role view treats mechanisms as functional roles that can be realized by different components. “Name mover” is a role: it describes what a component does (copy a name token to the output

position), not which head does it. The role view explains why functionally equivalent components across different models can be meaningfully compared.

Identity. Same functional input–output role, possibly in different components or models. Role identity requires a specification of the role that is independent of the discovery procedure (i.e., not merely “does what this head does”) and tests that go beyond the ablation that revealed the component.

Evidence. Role-specific predictions that are not entailed by mere causal relevance. For a name-mover role claim: the weight-space signature of W_{OV} should encode a copying operation; the head should perform this function on novel prompts; a head in a different model with the same W_{OV} signature should earn the same label. Tower [2026a] identifies the object-to-role slide as a common source of overclaiming: establishing that head 9.9 is causally necessary for IOI does not establish that it is a name mover until the role is independently operationalized and tested.

Failure modes. *Role proliferation*: a new role label is coined for each component discovered, re-describing the data without adding explanatory content. *Underdefined roles*: role labels are informal enough to be true of any component, making them unfalsifiable.

Discriminating experiment. Take the role specification and apply it to a model in which the original component has been ablated. If the role is real, either another component should take over (backup mechanism) or the behavior should degrade in a specific role-consistent way. If no such prediction is confirmed, the role label has not been independently validated.

3.3 Subspace View

The subspace view treats mechanisms as low-dimensional causal subspaces of the residual stream. The key observation is that distributed alignment search [Geiger et al., 2021, 2024] identifies not a component but a rotation of the activation space: the causal variable is encoded in a subspace that may span many components. If alignment matrix Q and its rotation QR both achieve the same interchange intervention accuracy (IIA), the relevant object is the subspace $\text{span}(Q)$, not the matrix Q .

Identity. Two subspace mechanisms are the same iff they project onto the same subspace of $\mathbb{R}^d$: same projector $QQ^\top$, or equivalently small geodesic distance on $\text{Gr}(k, d)$. This is basis-invariant and can be meaningful cross-model when corresponding residual streams can be aligned.

Evidence. DAS and IIA, subject to a critical constraint. Sutter et al. [2025] showed that unrestricted nonlinear alignment maps achieve 100% IIA even on randomly initialized models, making DAS vacuous without structural constraints on the alignment class. Subspace-view claims therefore require either linear alignment or alignment constrained to respect the transport structure induced by the weight matrices. Subject to this constraint, subspace stability (eigenvalue gap between the causal subspace and its complement) is the natural diagnostic.

Failure modes. *SAE confusion*: sparse autoencoder features are dictionary elements that minimize reconstruction loss; causal subspaces are directions that mediate specific causal paths. These are different objectives and can disagree. An SAE feature is a hypothesis about a one-dimensional subspace mechanism, not a confirmed subspace claim. *Sutter vacuousness*: reporting high IIA without specifying the alignment class allows any model to be assigned any causal structure.

Discriminating experiment. Verify that the identified subspace is stable: the subspace recovered by DAS on a held-out distribution should have small geodesic distance from the training-distribution subspace. Also confirm that an aligned basis-change (within the subspace) does not change the IIA, while a rotation that takes vectors out of the subspace does reduce it.

3.4 Structural View

The structural view treats mechanisms as equivalence classes of implementations under transformations that preserve the computation. The relevant object is not a head index, a basis vector, or a specific projection—it is the orbit of implementations related by the symmetry group of the transformer’s weight space. Two weight configurations that implement the same computation via different parameterizations are the same mechanism under this view.

The formalism involves gauge orbits, parallel transport maps on the Grassmannian, holonomy groups, and sheaf cohomology, developed in Tower [2026b].

Identity. Gauge-orbit membership: $x \sim y$ iff there exists a computation-preserving symmetry g (such as a scale transformation, a rotation of a head’s query-key subspace paired with an inverse rotation of its output, or a permutation of neurons) such that $g \cdot x = y$. Holonomy fingerprints provide a finer criterion: the holonomy group of the transport map induced by a mechanism’s weight matrices is an invariant that does not change under gauge transformations.

Evidence. Gauge-invariant measurements: weight-space composition scores $\|W_{O,u}W_{K,v}\|_F$ measure information flow capacity independently of basis choice [Tower, 2026c]; holonomy measurements; and reparameterization-stability tests (check that a result survives computation-preserving rescalings of the weights).

Failure modes. *False gauge invariance*: claiming structural equivalence without verifying that the symmetry actually preserves the computation. *Architectural prior confound*: structural features that discriminate trained from randomly initialized models may be detecting architecturally privileged positions rather than learned structure [Tower, 2026c].

Discriminating experiment. Apply a known computation-preserving symmetry (e.g., scale by λ , counter-scale by λ^{-1}) to the weights of a component and retest the activation-space evidence. If the activation-space result changes, the original result was not gauge-invariant and is not a structural-view claim.

3.5 Process View

The process view treats mechanisms as temporally extended. A mechanism is not merely a final-checkpoint artifact but a formation process, an execution trajectory, or a dynamical regime. This view is natural for phenomena where the how of learning is itself mechanistically relevant: the induction head mechanism, for instance, forms via a phase transition at a specific training step [Olsson et al., 2022]—this is process-level structure that is not recoverable from the final checkpoint alone.

Identity. Same trajectory type or dynamical basin. Two processes are the same mechanism if they follow the same formation pathway, occupy the same attractor, or exhibit the same phase-transition structure. This notion of identity is less developed than component or subspace identity, and formalizing it is an open problem.

Evidence. Checkpoints, gradient flow, loss-curve structure, and formation knockouts. A mechanism claim under the process view cannot be supported by final-checkpoint analysis alone. The induction head formation result provides a clear example: the phase change at a specific training step is visible in the composition score evolution, and this is process-level evidence distinct from any claim about the final circuit [Olsson et al., 2022].

Failure modes. *Process–circuit conflation*: the final-checkpoint circuit and the formation process are distinct objects; the circuit does not determine the process. Two models may arrive at identical final circuits via very different training trajectories. *Training-detail overfitting*: a process described precisely enough to be informative may be specific to a particular optimizer, learning rate schedule, or dataset.

Discriminating experiment. Run the same architecture with different training hyperparameters (learning rate, batch size, optimizer) and compare the formation dynamics. If the final circuit is similar but the formation trajectory differs, the process-level claim is hyperparameter-sensitive and should be stated at the appropriate level of generality.

3.6 Stratified View

The stratified view treats mechanism space as organized into strata: point-localized objects, one-dimensional features, higher-dimensional subspaces, nonlinear manifolds, and fully distributed structures. Rather than forcing a binary choice between localized and distributed, the stratified view asks which stratum the evidence supports and whether the appropriate description changes with measurement resolution.

Figure 2 illustrates the key distinctions schematically. The natural formalism is Whitney stratification: a decomposition of mechanism space into smooth manifold strata that are partially ordered by containment of their closures. A mechanism’s stratum is determined by its dimensionality, its localizability in specific components, and the topology of its support in the residual stream.

Identity. Same stratum and same local equivalence class within the stratum. Two descriptions are the same mechanism if they name the same stratum and agree on local structure within that stratum.

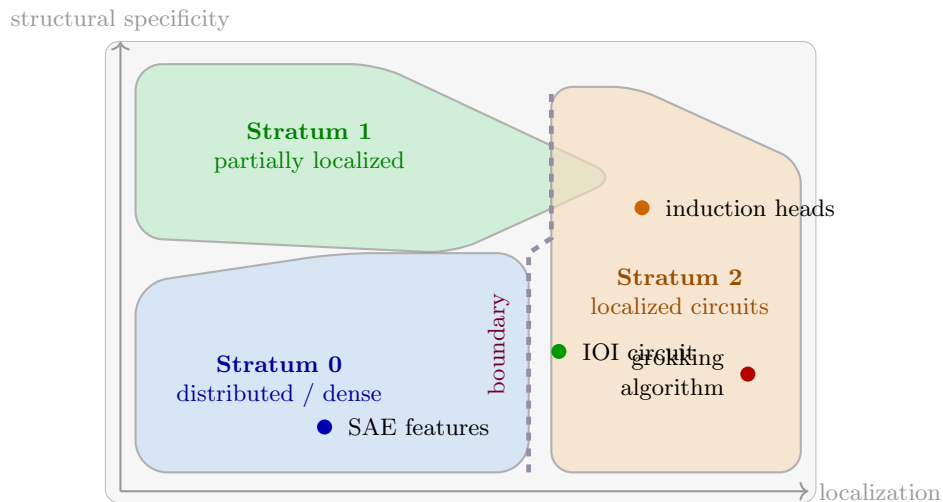


Figure 2: Schematic of mechanism space as a stratified manifold. Each stratum is a smoothly varying subset distinguished by the localization and structural specificity of the mechanism. Boundaries between strata are lower-dimensional singular sets. Empirical examples are placed roughly by the evidence reviewed in Section 4; exact positions are schematic, not quantitative.

Evidence. Dimensionality tests (effective rank of the causal subspace), localization diagnostics (is the mechanism’s effect concentrated in a small component set or distributed?), and stratum-transition tests (does the apparent mechanism change stratum when the measurement resolution changes?).

Failure modes. *Stratum misidentification from method artifacts.* A genuinely distributed mechanism may appear localized to a single component if the ablation method captures only the dominant direction. Conversely, a localized mechanism may appear distributed if the measurement method has high background noise. The stratified view does not resolve this automatically—it provides the language to state it precisely.

Discriminating experiment. The stratified view predicts that measurement resolution affects the apparent stratum. Test this by comparing the effective dimensionality of the causal subspace recovered by DAS at different hook points and with different alignment constraints. Consistent stratum across resolutions supports a stratum assignment; stratum-change suggests either a higher-stratum mechanism being partially recovered or a measurement artifact.

3.7 Perspectival View

The perspectival view treats methods as revealing partial projections of mechanisms rather than the mechanism in itself. Activation patching, weight analysis, DAS, and sparse autoencoders are different perspectives with different invariances and different blind spots. Disagreement between methods is therefore not automatically a failure; it may reveal the method-relative structure of what is being measured.

This view is philosophically related to Massimi’s perspectival realism [Massimi, 2022]: the claim is not that mechanisms are relative to methods (that would be instrumentalism), but

that scientific methods are partial perspectives on a method-independent mechanistic structure. Mechanistic claims are not merely predictions—they have a realist reading—but each method can only support claims about the aspect of the mechanism it is sensitive to.

Identity. Coherence across perspectives. Two descriptions refer to the same mechanism if they are mutually consistent and jointly interpretable as projections of the same underlying structure. This is weaker than convergence (the descriptions need not agree quantitatively) but stronger than mere non-contradiction.

Evidence. Multi-method robustness. A mechanistic claim supported by a single method is vulnerable to that method’s specific confounds; a claim supported by convergent evidence from methods with different assumptions is stronger. MechVal [Tower, 2026a] formalizes this as a convergence requirement across three evidence families: structural (weight), causal (activation/intervention), and dynamic (training-time).

Failure modes. *Perspectival conflation*: treating a method-specific disagreement as evidence against the existence of the mechanism, rather than as information about which aspect each method is tracking. *Anti-realist slide*: treating the method-relativity of evidence as implying that mechanisms are method-relative objects—conflating the perspectivity of evidence with the perspectivity of truth.

Discriminating experiment. Take a mechanistic claim supported by patching and test whether the weight-space composition score for the same component is consistent with the claimed causal role. If the weight-space result is inconsistent, this is either a perspectival disagreement (the two methods track different aspects of the structure) or a failure of one method. Adjudicating which requires a third method.

3.8 Instrumental View

The instrumental view treats mechanisms as useful predictive or interventional models. On this view, a mechanism claim is warranted if it helps predict or control the model, regardless of whether it identifies a real internal object. The label “head 9.9 is a name mover” is warranted if it generates accurate behavioral predictions and effective interventions, not because the head has an intrinsic name-moving property.

Identity. Predictive equivalence. Two descriptions are the same mechanism if they make the same predictions about all interventions and all relevant behaviors. This is behaviorism at the level of mechanism descriptions.

Evidence. Forecast accuracy and intervention utility. A mechanism label is justified to the extent that novel behaviors predicted from it are confirmed and interventions derived from it have the expected effects.

Failure modes. The instrumental view has the weakest ontological commitments and is therefore the easiest to satisfy—but this is also its primary limitation. *Safety-relevant overreach*: claims that require strong ontological conclusions (“the model is pursuing a goal”, “this behavior would persist under distribution shift”) cannot be supported by predictive utility on the training distribution alone. A mechanism that predicts behavior on seen distributions may fail on unseen ones in ways that an object- or structure-level account would anticipate.

Discriminating experiment. Test whether the mechanism label generates predictions that go beyond the behavior used to identify it. If a head is labeled a name mover, it should predict name-moving behavior in novel syntactic constructions not present in the original test suite. Failure of novel prediction at the same rate as success on the original test is evidence for an overfitted label.

4 Worked Examples

Three extended examples show what the atlas does in practice: not just describing views in the abstract, but showing how the same body of evidence is read differently under different views and what each reading implies for follow-up work.

4.1 Indirect Object Identification

Wang et al. [2022] identify a circuit for indirect object identification (IOI) in GPT-2 Small: 26 attention heads organized into functional classes including S-inhibition heads, name movers, and duplicate-token detection heads.

Object view. The mechanism is those 26 heads. The evidence (activation and path patching, mean ablation) establishes causal necessity and sufficiency relative to the test distribution. Object-level identity is precise within GPT-2 Small and fails across architectures.

The object-level result is solid. What it does not establish: (a) that these heads perform a functional role that would be recognizable in another model, (b) that the relevant causal variables are localized in specific subspaces, (c) that the result is invariant under parameter reparameterizations, or (d) how the circuit formed. Claims (a)–(d) require distinct evidence.

Role view. The four role classes (S-inhibition, name mover, duplicate-token detection, and backup name mover) constitute a role-level description. This is a hypothesis about what the heads do, not a confirmed role claim. To confirm it, each role should be operationalized and tested independently: name movers should copy tokens in novel syntactic constructions; S-inhibition heads should suppress the subject in novel contexts where the subject–object relationship is structurally varied. Until these tests are run, the role labels describe the experimental template more than the mechanism.

Subspace view. The subspace question is where the IO and S tokens, and the inhibitory relationship between them, are represented in the residual stream. DAS with linear alignment constraints would identify the causal subspaces for each variable and test whether they can be independently intervened on. The 26-head object-level description does not resolve this: two different subspace configurations could produce the same patching results.

Structural view. The structural question is which aspects of the circuit are gauge-invariant. Path-patching results that survive computation-preserving reparameterizations of the relevant head weights establish structural facts; those that do not are basis-dependent observations. The holonomy of the circuit’s transport map would provide a structural fingerprint that is invariant across model instances.

What each view implies for next steps. Under the object view, the main open question is distribution robustness: does the 26-head circuit generalize to other indirect object constructions beyond the original template? Under the role view, it is role operationalization: what behavioral predictions do the role labels generate beyond the original task? Under the subspace view, it is subspace identification and stability. Under the structural view, it is gauge-invariant characterization via the companion geometric formalism.

4.2 Induction Heads

The induction head result [Olsson et al., 2022] is the strongest mechanistic result in the current literature, precisely because multiple views converge on the same object.

Object view. Induction heads are identified by activation patching. Their distinctive attention pattern—attending to the token immediately after the previous occurrence of the current token—is established across a large family of transformers. Object-level identity is extended cross-model by demonstrating the pattern holds across model sizes and architectures.

Role view. The induction pattern defines a functional role that is operationalized (it can be tested on any sequence with a repeated token) and confirmed across model families. This is one of the few cases in the literature where role identity is genuinely established rather than hypothesized. The role specification is independent of the component: any head that attends to t_j from position i when token $t_i = t_{j-1}$ is an induction head, regardless of layer or position.

Process view. The formation dynamics are studied across training checkpoints. A phase transition is identified at a specific step in the loss curve, and the composition score between the head and its prefix-matching partner increases sharply at that step [Olsson et al., 2022]. This is process-level evidence—it concerns how the mechanism formed, not what it is at the final checkpoint. The formation process is argued to be causally responsible for the in-context learning capacity of the model, a claim that requires process evidence precisely because it is about the training-time origin of a behavior.

What remains open. Under the stratified view: induction heads appear to be point-localized objects, but whether the relevant causal variable is localized to a one-dimensional subspace (object stratum) or spans a small-dimensional subspace is not established. Under the structural view: the holonomy of the induction head transport map has not been characterized. Under the subspace view: the exact subspace of the residual stream encoding the “previous occurrence” variable and the “current token” variable has not been jointly identified via DAS.

The convergence across object, role, and process views is why the induction head result can support relatively strong claims. The gaps in subspace and structural evidence are where further work would strengthen or qualify those claims.

4.3 Sparse Autoencoder Features

Sparse autoencoder (SAE) features recovered from Claude-3 Sonnet [Templeton et al., 2024] and smaller models [Bricken et al., 2023] are semantically interpretable and partially causally validated via steering experiments. The status of SAE features as mechanisms is genuinely contested, and the atlas provides precise vocabulary for the dispute.

Object view. Under the object view, each dictionary element is a potential mechanism: a one-dimensional object in the activation space. The evidence (semantic interpretability, steering effects) supports a weak form of causal relevance. The object-level description is precise: the feature is the specific dictionary vector learned by the SAE.

Subspace view. Each feature is a hypothesis about a one-dimensional causal subspace mechanism. The hypothesis is that the direction of the feature vector corresponds to a direction in the residual stream that mediates a specific causal variable. The SAE training objective (sparse reconstruction) does not guarantee this: the feature might be the dominant activation direction without being the causally relevant one. Confirmation requires linear DAS showing that the feature direction specifically mediates the causal variable in question.

Perspectival view. SAE features are a measurement perspective on the activation space: the perspective defined by the sparse coding objective. This perspective has specific invariances (insensitive to causal structure; sensitive to geometric structure of activations) and specific blind spots (may miss subspaces that are causally relevant but not reconstructively prominent). Under the perspectival view, the right question is not “are SAE features mechanisms?” but “what does the SAE perspective reveal that the weight-space perspective and the causal intervention perspective would independently confirm?” Convergence of all three on the same direction would substantially strengthen the claim.

What each view implies for next steps. Under the object view: test causal necessity via ablation and compare with the steering evidence. Under the subspace view: run linear DAS for the most interpretable features to test whether the feature direction is the causal subspace direction for the relevant variable. Under the perspectival view: compute weight-space composition scores for the same features to identify which ones are also structurally prominent, and compare with the causal evidence.

5 Five Decision Points

Five recurring forks in empirical interpretability work change in meaning depending on which view is operative. We examine what each view implies for each fork and, where views make conflicting predictions, what evidence would adjudicate.

5.1 Localized vs. Distributed

The question is whether a mechanism is implemented by a small, specific set of components or spread across many.

Under the *object view*, this is an empirical question: which minimal component set is causally necessary and sufficient? Under the *subspace view*, it is about the dimensionality and localizability of the causal subspace: a low-rank subspace localized in a small number of attention heads is one answer; a high-dimensional subspace spanning many heads is another. Under the *stratified view*, it is a question about which stratum the evidence supports.

The views come apart in practice. Activation patching may identify a small object-level component set as causally necessary, while DAS identifies a high-dimensional subspace spanning many components. These need not be contradictory: the patching identifies the components whose activation values lie in the relevant subspace; DAS characterizes the subspace. The stratified view provides the language to say that the two results are compatible descriptions at different levels of resolution.

5.2 Object vs. Role

The question is whether a mechanism claim is about a specific component or about a functional role that component plays. Under the *object view*, the claim is about the component; under the *role view*, it is about the function.

The distinction matters for generalization. An object claim does not entail that any component in any other model is the same mechanism. A role claim does—if the role is operationalized and tested. The slide from object to role happens in one sentence in most papers (“head 9.9, which we call a name mover, ...”) and is a common source of overclaiming identified in MechVal [Tower, 2026a]: the object identity is established, the role is named, and subsequent sentences treat the role as established.

5.3 Single vs. Triangulated Evidence

The question is whether a mechanistic claim requires evidence from a single method or convergent evidence from multiple methods with independent assumptions.

Under the *object view* with activation patching, a single strong patching result may be sufficient. Under the *subspace view*, weight-space and activation-space convergence strengthens the claim (both should identify the same subspace if the view is correct). Under the *perspectival view*, single-method evidence is always vulnerable to that method’s confounds: triangulation is definitionally required.

The case for triangulation is not merely a caution. If a method m can only support claims in its range $C(m)$, and the target claim is not in $C(m)$, then no amount of evidence from m can support the claim—however strong that evidence is.

5.4 Static vs. Process

The question is whether the mechanism is a final-checkpoint artifact or a formation process. This is a clean view-level distinction with a clean evidential consequence: process claims need process evidence; checkpoint claims cannot support process conclusions.

This matters most for: (a) training-dynamics hypotheses (“the model learned to do X because ...”), (b) phase-transition-like generalization patterns, and (c) claims about the origin of in-context learning behaviors. In all three cases, checkpoint evidence establishes what the mechanism is at convergence but not how it came to be.

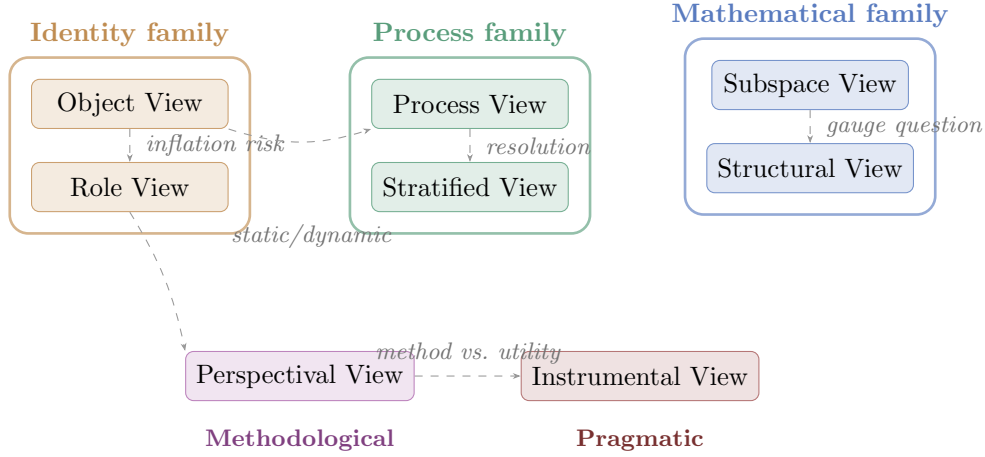


Figure 3: The eight views grouped into families by shared mathematical apparatus (identity, process, mathematical), with two singletons (methodological, pragmatic). Dashed arrows mark principal decision points and conflation risks discussed in Section 5.

5.5 Subspace vs. Structural

The question is whether mechanism identity is determined by the subspace a mechanism occupies or by the gauge-invariant structure of the computation it performs. A subspace claim is basis-invariant: the subspace $\text{span}(Q)$ is the same regardless of which orthonormal basis of Q is used. But it is not symmetry-invariant: two subspace mechanisms in the same gauge orbit (related by a computation-preserving symmetry that rotates the subspace) would be the same mechanism under the structural view but potentially different under the subspace view.

The structural view is more realist and harder to operationalize. The subspace view is more tractable and directly supported by current DAS methodology. For most practical questions in the field, the subspace view is appropriate; the structural view becomes important when comparing across parameterizations of the same architecture or when making claims that should be invariant to training randomness.

6 Relation to Companion Frameworks

Three companion papers address complementary parts of the problem. We describe each and clarify what it presupposes and what it provides.

Mechanistic Validity [Tower, 2026a]. MechVal provides 27 criteria for evaluating whether a mechanistic interpretability claim is warranted by evidence. The criteria are organized across five validity lenses (construct, internal, external, measurement, interpretive) and a tier taxonomy (Speculative through Established).

MechVal presupposes a claim type: it evaluates whether the evidence supports the claim, not whether the claim is well-formed. Mechanistic Views is logically prior. The MechVal lens for interpretive validity includes a level-declaration criterion (V1) requiring every quantitative claim to be labeled with its level of analysis. Mechanistic Views provides the vocabulary for those labels: an activation-patching result is an object-level observation; a DAS result (subject to the

Sutter constraint) is a subspace-level observation; a holonomy measurement is a structural-level invariant.

The Geometry of Mechanisms [Tower, 2026b]. This paper provides the formal mathematical objects that the subspace and structural views implicitly require: Grassmannian structural causal models (G-SCMs), parallel transport maps between Grassmannians, holonomy groups, and sheaf cohomology organized over the circuit graph. The G-SCM makes the subspace view precise: mechanisms are subspace-valued causal variables, and evidence is organized into three functors ($\mathcal{E}_{\text{struct}}$, $\mathcal{E}_{\text{causal}}$, $\mathcal{E}_{\text{dynamic}}$).

The paper also develops a conjecture about the injectivity of joint evidence:

Conjecture 6.1 (Evidence injectivity; Tower 2026b). *Each of the three evidence functors $\mathcal{E}_{\text{struct}}$, $\mathcal{E}_{\text{causal}}$, and $\mathcal{E}_{\text{dynamic}}$ fails to identify mechanisms that the other two can distinguish. The joint evidence functor $\mathcal{E} = (\mathcal{E}_{\text{struct}}, \mathcal{E}_{\text{causal}}, \mathcal{E}_{\text{dynamic}})$ is injective on the space of natural transformer mechanisms.*

We state this as a conjecture rather than a theorem because the companion paper is a working draft and the result requires verification. The claim that single-method evidence is insufficient for mechanism identity is motivated by Proposition 2.1 and the Sutter result; the claim that joint evidence is sufficient is an empirical conjecture about the space of transformer mechanisms.

Weight-Space Circuit Analysis [Tower, 2026c]. This paper provides empirical grounding for the structural view via analysis of 123 geometric features extracted from static weight matrices. The features recover known circuits at 84–100% edge F1 with zero forward passes, achieving weight–activation IIA correlation $r = 0.982$ across 432 head–task pairs.

The weight-space results are the natural evidence for the structural view: they are distribution-free, defined before any input is shown to the model, and invariant to prompt distribution and sampling. The composition score $\|W_{O,u}W_{K,v}\|_F$ provides a distribution-free upper bound on information flow that is a structural claim inaccessible to activation-based methods.

7 Implications and Open Questions

Stating the view is a publication norm. The primary practical implication of this paper is that mechanistic interpretability claims should state the view under which they are made. This is not bureaucratic overhead—it is a precision requirement of the same kind as reporting which distribution patching was run on. Object-level claims should be labeled as such, with the relevant distribution specified. Role claims should specify the role operationally and report the independent tests. Subspace claims should report the identity criterion (same projector? geodesic distance below what threshold?) and the subspace stability. Structural claims should report which symmetries were tested. Process claims should report the temporal evidence.

Evidence-claim matching. Each measurement type has a range: the set of claim types it can support. A measurement whose range does not include the target claim type cannot support that claim, regardless of how strong the measurement is. Activation patching is not in

the range of structural claims. Single-method IIA without alignment constraint is not in the range of any claim. DAS with linear alignment is in the range of one-dimensional subspace claims. Stating the view makes these range requirements explicit and checkable.

Views as research programs. The object and role views primarily organize and classify existing empirical practice. The subspace, structural, and stratified views generate positive research programs: each calls for methods, results, and formalism that do not yet exist. For the subspace view: systematic DAS studies with stability tests across models and distributions. For the structural view: holonomy measurements, gauge-invariance checks, and the development of the sheaf cohomology formalism. For the stratified view: dimensionality diagnostics and stratum-transition tests. The atlas is therefore not merely a taxonomy of current practice; it is a map of what is missing.

Open questions. Several questions are raised by the framework and not resolved:

1. *Process identity.* The identity criterion for the process view (same dynamical basin or trajectory type) is underspecified relative to the object, subspace, and structural criteria. Formalizing process identity is an open problem.
2. *Cross-view translation.* Under what conditions can an object-level result be promoted to a subspace-level or structural-level result? What additional evidence is required? A systematic account of inter-view inference is missing.
3. *Completeness of the atlas.* The eight views are identified by reading the literature, not derived from the five axes. Whether there are important views not represented—a Marr-level-2 computational view, a representational-content view, a population-level statistical view—is an open question.
4. *View selection.* The framework is pluralist but does not say which view to adopt for a given question. A principled account of view selection—given a target phenomenon T , which view(s) are appropriate?—would extend the framework substantially.

8 Conclusion

Mechanistic interpretability is not only a search for circuits. It is a search for the right kind of object to call a mechanism. The field already uses multiple mechanistic views, typically implicitly, and making those views explicit improves both the precision of claims and the appropriateness of evidence.

The central contribution is definitional: a mechanistic view is a five-component structure $(O, \sim, E, F, T)$, and every mechanistic interpretability result assumes one, whether or not it is stated. Coherence between the components is not automatic, and incoherent views generate systematically unsatisfiable evidential demands.

The atlas of eight views—object, role, subspace, structural, process, stratified, perspectival, instrumental—covers the main options currently in use. Each view is appropriate for some questions and inappropriate for others. The goal is not to collapse this plurality into a master ontology but to make view choice explicit, justifiable, and revisable. The induction head result remains the canonical illustration of what convergence across views looks like: object, role,

and process views all point to the same entity, which is why the result supports claims that single-view results cannot.

Acknowledgments

The MechVal framework, the geometric formalism, and the weight-space experiments are developed in companion working papers. All errors are the author’s own.

A Coherence Violations: A Diagnostic Checklist

The following patterns indicate a coherence violation in the view underlying a claim.

- (1) **Ontology–identity mismatch.** The ontology declares mechanisms to be concrete components, but the identity criterion is functional (two different components can be the same mechanism). Resolution: adopt the role view, which has functional identity built in, or restate the claim as object-level and drop cross-model generalization.
- (2) **Evidence out of range.** A structural or subspace claim is supported only by activation-patching evidence. Resolution: identify the view that patching evidence actually supports (object view) and state the structural or subspace claim as a hypothesis requiring further evidence.
- (3) **Unrestricted alignment.** A subspace claim is supported by DAS/IIA without specifying the alignment class or reporting subspace stability. By the Sutter result [Sutter et al., 2025], high IIA under unrestricted nonlinear alignment is consistent with any causal structure. Resolution: restrict the alignment class to linear maps or transport-respecting maps and report the eigenvalue gap.
- (4) **Process claim from checkpoint evidence.** A claim about how a mechanism formed or why a behavior emerged at a certain scale is supported only by final-checkpoint analysis. Resolution: either restate as a static claim or obtain training-dynamics evidence.
- (5) **Object-to-role slide.** A component is identified by ablation and immediately labeled with a role without an independent test of the role specification. Resolution: operationalize the role prediction and test it on held-out constructions.

B Full Axis Assignments

Table 2 repeats the five-axis assignments from Table 1 and adds the characteristic failure mode for each view.

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View	Ontology	Identity	Evidence	Formalism	Target	Failure mode
Object	Concrete part	Component overlap	Ablation, patching	Directed graph	Specific behavior	Role inflation; distribution sensitivity
Role	Functional role	Role equivalence	Role-specific causal tests	Functional decomp.	Functional class	Role proliferation; underdefined roles
Subspace	Causal subspace	Same projector	DAS/IIA (linear)	$\text{Gr}(k, d)$	Representations variable	Slutter vacuousness; SAE conflation
Structural	Gauge-invariant structure	Gauge orbit	Holonomy, composition	Fiber bundle	Computation class	False gauge invariance; arch. prior confound
Process	Formation trajectory	Trajectory type	Checkpoints, knockouts	Dynamical system	Mechanism origin	Process-circuit conflation; training overfitting
Stratified	Stratum point	Stratum + local equiv.	Dimensionality, rank	Whitney strat.	Resolution-relative	Stratum misidentification from method artifacts
Perspectival	Method projection	pro- coherence	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative
Instrumental	Predictive model	Predictive equiv.	Forecast, intervention	Model theory	Behavior prediction	Safety-relevant overreach

Table 2: Full axis assignments and characteristic failure modes.

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